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MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11732360 SUPERELEVATION AND DRIFT

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly: 11732360 Superelevation and Drift which is the A5 component of the Ballistic Computer Unit, 11732400.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-STD-45662	Calibration System Requirements

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11732360	Circuit Card Assembly: 11732360 Superelevation and Drift

Inspection Equipment

11750202	Automatic Card and Unit, Test Set
11750422	Adapter, Computer Board-A5
11821457	Fixture Shock, Vertical and Horizontal Plane
11821458	Shock Adapter, Plate Vib Horz-Vert Plane/ Shock-Z
11821500	Vibration/Shock Test Fixture-Computer Unit, FAT
11821520	Automatic Test Assembly - Central Processor Unit
11921854	Test Fixture Auto Card, Test Set, Computer FAT

Packaging Data Sheet

P11732360	Packaging of Circuit Card Assembly: 11732360

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The superelevation and drift circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11732360 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Signals. The assembly shall provide the following signals:

Signal Designation	Signal Nomenclature
a. R_o(AC)	Target range
b. R_oP BAR	Normalized air pressure
c. $-R_oT_A$ BAR	Normalized air temperature
d. $-R_o$ WEAR BAR	Muzzle velocity correction
e. Epsilon	Superelevation
f. Eta	Azimuth lead

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a. Temperature	73° ± 18°F
b. Relative humidity	50 percent + 30 percent −3.0
c. Atmospheric pressure	28.5 + 2.0 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

3.3.1 R_o(AC) output. With the load (item 1.1 of table I) connected, the power sources (items 2.1 and 2.2 of table I) applied and the input V_{5A}, (0 to 5 vrms, phase 1) applied at P1-5A, the nominal output of P1-25A shall be described by the following transfer function:

$$V_{25A} = 1.4100 * V_{5A}$$

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TABLE 1. Loads, power, signal and logical sources.

Item	Conditions	Characteristics	Connections
1.0	Load		Connected between the following points:
1.1	Resistor	1000 ohms ± 5%, 1/4 W	P1-25A and P1-13A
2.0	Power sources		Applied between the following pins and P1-13A (rtn):
2.1	15.000 vdc (+15V)	Tolerance: ±0.150 vdc Ripple: 20 mv p-p max	P1-6A
2.2	-15.000 vdc (-15V)	Tolerance: ±0.150 vdc Ripple: 20 mv p-p max	P1-21A
2.3	10.000 vdc (+10V)	Tolerance: ±0.015 vdc Ripple: 5 mv p-p max	P1-4B
2.4	-10.000 vdc (-10V)	- item 2.3 ±0.010 vdc Ripple: 5 mv p-p max	P1-25B
3.0	Signal sources	As specified below: 1/	Applied as specified with return to P1-13A:
3.1	400 Hz square wave (MR1)	See Figure 1.	
4.0	Logical sources	As specified below 2/	Applied between the following pins and P1-13A (rtn):
4.1	AS1	(Logical 1) 0.70 ± 0.50 vdc in series with 100 ohms ± 5% resistor (Logical 0) -15.0 ± 0.5 vdc in series with 51K ohms ± 5% resistor	P1-1A
4.2	AS2	Same characteristics as item 4.1	P1-2B

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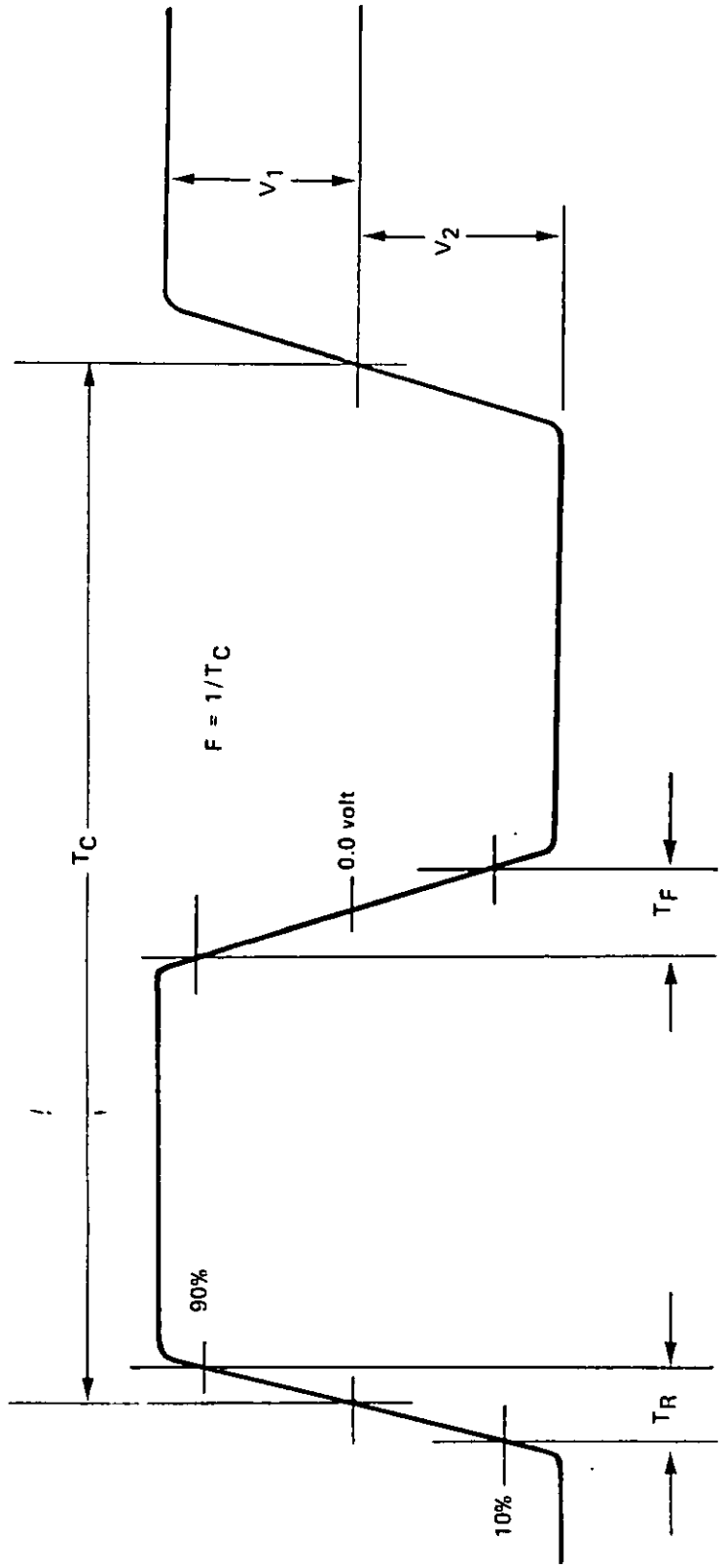
TABLE I. Loads, power, signal and logical sources. - Continued

Item	Conditions	Characteristics	Connections
4.3	AS3	Same characteristics as item 4.1	P1-2A
4.4	AS4	Same characteristics as item 4.1	P1-1B
4.5	AS5	Logical 0 of item 4.1	P1-3B
4.6	AS6	Same characteristics as item 4.1	P1-5B
4.7	BIT	(Logical 1) 13.0 ± 0.5 vdc in series with 510 ohms ± 5% resistor (Logical $\overline{0}$) -14.5 ± 0.5 vdc in series with 10K ohms ± 5% resistor	P1-22B
4.8	$\overline{\text{BIT}}$ (BIT BAR)	Logical complement of item 4.7 with same characteristics.	P1-21B

1/ a) Positive (+) magnitudes (phase 1 signals) shall be in phase ± 1.0° with item 3.1
b) Negative (-) magnitudes (phase 2 signals) shall be 180° ± 1.0° out of phase with item 3.1

2/ Logical terms have a value of either one (1) or zero (0) as defined by the voltage levels in the characteristics column.

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	Standard ambient	140°F	-25°F
V_1	15± 0.50 volts	15± 0.50 volts	15± 0.50 volts
V_2	- 15± 0.50 volts	- 15± 0.50 volts	- 15± 0.50 volts
T_R	0.5 to 5 microsecond	0.5 to 4 microsecond	1 to 6 microsecond
T_F	0.5 to 4 microsecond	0.5 to 4 microsecond	0.5 to 6 microsecond
F	400± 1 Hz	402± 2 Hz	398± 2 Hz

FIGURE 1 MR1 waveform.

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The tolerance of the output shall be not greater than the value of the following equation:

$$\Delta V_{25A} = 0.0004 * |V_{5A}| + 0.001$$

Application of the transfer function and the tolerance equation shall be valid within the following limit:

$$|V_{25A}| < 7 \text{ vrms} \qquad \text{(See 4.6.2.1)}$$

3.3.2 $R_O\bar{P}$ output. With the load (item 1.1 of table I) connected, power sources (items 2.1 and 2.2 of table I) applied and the inputs (table II) applied, the nominal output of P1-9A shall be described by the following transfer function:

$$V_{9A} = 1.300 * V_{10B} - 1.300 * V_{10A}$$

TABLE II. $R_O\bar{P}$ inputs.

Item	Input Signal	Pin Location 1/	Signal Limits
1	V_{10A}	P1-10A	0 to 7 vrms, phase 1 (+) or phase 2 (-)
2	V_{10B}	P1-10B	Same as item 1

1/ Returns to P1-13A.

The tolerance of the output shall be not greater than value of the following equation:

$$\Delta V_{9A} = 0.013 * |V_{10B}| + 0.013 * |V_{10A}| + 0.001$$

Application of the transfer function and the tolerance equation shall be valid within the following limit:

$$|V_{9A}| < 6 \text{ vrms} \qquad \text{(See 4.6.2.2)}$$

3.3.3 $-R_O\bar{T}_A$ output. With the load (item 1.1 of table I) connected, power sources (items 2.1 and 2.2 of table I) applied and the inputs (table III) applied, the nominal output of P1-12A shall be described by the following transfer function:

$$V_{12A} = 1.470 * V_{14A} - 1.470 * V_{12B}$$

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TABLE III. $-R_oT_A$ inputs.

Item	Input Signal	Pin Location 1/	Signal Limits
1	V_{14A}	P1-14A	0 to 7 vrms, phase 1(+) or phase 2(-)
2	V_{12B}	P1-12B	Same as item 1

1/ Returns to P1-13A.

The tolerance of the output shall be not greater than the value of the following equation:

$$\Delta V_{12A} = 0.0147 * \left|V_{14A}\right| + 0.0147 * \left|V_{12B}\right| + 0.001$$

Application of the transfer function and the tolerance equation shall be valid within the following limit:

$$\left|V_{12A}\right| < 6 \text{ vrms} \qquad \text{(See 4.6.2.3)}$$

3.3.4 $R_o\overline{WEAR}$ output. With the load (item 1.1 of table I) connected, power sources (items 2.1 and 2.2 of table I) applied and the input V_{20A} (0 to 7 vrms, phase 1(+) or phase 2(-)) applied at P1-20A, the nominal output of P1-19A shall be described by the following transfer function:

$$V_{19A} = -1.000 * V_{20A}$$

The tolerance of the output shall be not greater than the value of the following equation:

$$\Delta V_{19A} = 0.010 * \left|V_{20A}\right| + 0.001$$

Application of the transfer function and the tolerance equation shall be valid within the following limit:

$$\left|V_{19A}\right| < 6 \text{ vrms} \qquad \text{(See 4.6.2.4)}$$

3.3.5 Epsilon output. With the load (item 1.1 of table I) connected, power sources and signal sources (items 2.1, 2.2, 2.3, 2.4, and 3.1 of table I) applied and the inputs (table IV) applied, the nominal output of P1-4A shall be described by the following transfer function:

$$V_{4A} = K_1 * R_A + K_2 * R_B + K_3 * R_C$$

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where:

a. Quantities R_A, R_B and R_C are calculated using the following equations:

1. $R_A = -1.4100 * V_{5A} - 0.5000 * R_V$

2. $R_V = K_4 * (1.4100 * V_{5A} * SW1 - V_{20A} * SW2) + 1.4100 * K5 * V_{5A}$

3. $R_B = 0.30 * \frac{10}{V_{REF}} * \left(\frac{V_{5A}}{1.0638} + \frac{R_V}{3} \right) * \frac{R_M}{1.5} + R_A * \frac{(V_{4B} + V_{25B})}{10}$

4. $R_C = -0.06 * \left(\frac{10}{V_{REF}} \right)^2 * \left(\frac{V_{5A}}{1.0638} + \frac{R_V}{3} \right) * \left(\frac{R_M}{1.5} \right)^2 + R_B * \frac{(V_{4B} + V_{25B})}{10}$

TABLE IV. Epsilon inputs.

Item	Input Signal	Pin Location 1/	Signal Limit
1	V_{25A}	P1-25A	See 3.3.1
2	V_{19A}	P1-19A	See 3.3.4
3	V_{9A}	P1-9A	See 3.3.2
4	V_{12A}	P1-12A	See 3.3.3

1/ Returns to P1-13A.

5. $R_M = -1.4100 * V_{5A} - 0.3150 * (1.3000 * V_{10B} - 1.3000 * V_{10A}) * SW2 - 1.4700 * 0.2403 * (V_{14A} - V_{12B}) * SW2 - 0.1016 * R_V$

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6. $V_{REF} \frac{V_{4B} - V_{25B}}{2}$

7. V_{4B} is defined in table 1, item 2.3.

8. V_{25B} is defined in table I, item 2.4.

b. The values of K_1, K_2, K_3, K_4 and K_5 are determined for each item from table V.

c. Terms SW1 and SW2 have the value of either 1 or 0 as defined by the logical expressions:

1. SW1 = B1T

2. SW2 = $\overline{B1T}$

The tolerance of the output shall be not greater than the value of the following equation:

$$\Delta V_{4A} = 0.0005 * |V_{4A}| + \Delta K_1 * |R_A| + \Delta K_2 * |R_B| + \Delta K_3 * |R_C| + |K_1| * \Delta R_A + |K_2| * \Delta R_B + |K_3| * \Delta R_C + 0.001$$

Where:

a. ΔR_A, ΔR_B and ΔR_C are calculated using the following equations:

1. $\Delta R_A = 0.0005 * (|R_A| + |1.4100 * V_{5A}| + |R_V|)$

2. $\Delta R_B = 0.002 * |R_B| + 0.002$

3. $\Delta R_C = 0.002 * |R_C| + 0.003$

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TABLE V. Epsilon and eta constants.

Item	Selected inputs 1/	K_1	K_2	K_3	K_4	K_5	K_6	ΔK_1	ΔK_2	ΔK_3
1	AS1	-0.151626	-0.021882	-0.06056	-0.0660	0.1268	0.0342	0.000349	0.000815	0.001479
2	AS1, AS6	-0.151626	-0.021882	-0.06056	-0.0660	0.0000	0.0342	0.000349	0.000815	0.001479
3	AS2, AS6	-0.259860	-0.054973	-0.46348	-0.0000	0.0000	0.0000	0.000432	0.000836	0.001471
4	AS3, AS6	-0.639960	-0.559067	-0.85530	-0.2040	0.0000	0.0366	0.000577	0.000941	0.001534
5	AS4, AS6	-0.148095	-0.021838	-0.00867	-0.0660	0.0000	0.0000	0.000345	0.000813	0.001248

1/ Selected inputs are set to logical 1. (See item 4.0 of table I)

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b. The values of ΔK_1, ΔK_2 and ΔK_3 are specified in table V.

Application of the transfer function and the tolerance equation shall be valid within the following limit:

$$|V_{4A}| < 7 \text{ vrms} \qquad \text{(See 4.6.2.5)}$$

3.3.6 Eta output. With the power sources (items 2.1 and 2.2 of table I) applied and the inputs (table VI) applied, the nominal output of P1-14B shall be described by the following transfer function:

$$V_{14B} = - K_6 * V_{4A} - 1.0000 * V_{18B}$$

where the value of K_6 is specified in table V.

TABLE VI. Eta inputs.

Item	Input Signal	Pin Location 1/	Signal Limits
1 2	V_{4A} V_{18B}	P1-4A P1-18B	See 3.3.5 0 to 7 vrms, phase 1

1/ Returns to P1-13A.

The tolerance of the output shall be not greater than the value of the following equation:

$$\Delta V_{14B} = 0.0005 * |V_{14B}| + 0.0005 * |V_{4A}| + 0.0005 * |V_{18B}| + 0.0015$$

Application of the transfer function and the tolerance equation shall be valid within the following limit:

$$|V_{14B}| < 7 \text{ vrms} \qquad \text{(See 4.6.2.6)}$$

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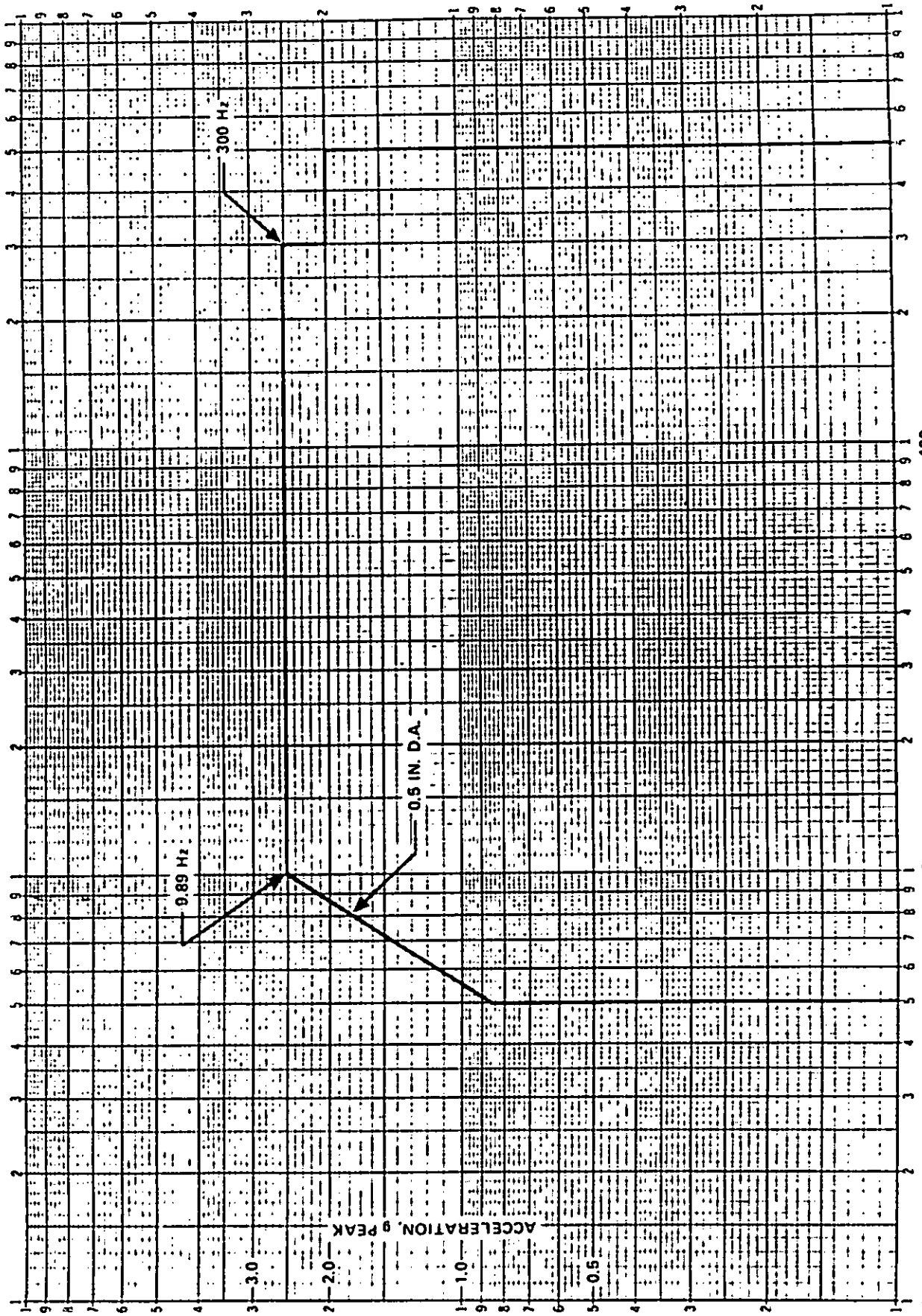


FIGURE 2. Vibration profile.

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3.4 Environmental.

3.4.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity units (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 2. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

3.4.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to +140°F. (See 4.6.3.3, 4.6.3.4, 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to +160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First Article inspection (see 4.4)
2. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-15926 and the conditions of 3.1.3.

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4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables VII, VIII and IX.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated resquirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

TABLE VII. Performance tests.

Item	Characteristic	Requirement	Test Procedure
101	R_O(AC) output	3.3.1	4.6.2.1
102	$R_O\overline{P}$ output	3.3.2	4.6.2.2
103	$-R_O\overline{T}_A$ output	3.3.3	4.6.2.3
104	$-R_O\overline{WEAR}$ output	3.3.4	4.6.2.4
105	Epsilon output	3.3.5	4.6.2.5
106	Eta output	3.3.6	4.6.2.6

TABLE VIII. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - Visual
302	General Specification	3.1.2	MIL-F-13926 - Visual

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TABLE IX. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.4.1	4.6.3.1
304	Vibration	3.4.2	4.6.3.2
305	High Temperature	3.4.3 and 3.4.4	4.6.3.3
306	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
307	Temperature Cycling	3.4.3 and 3.4.4	4.6.3.5

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table VIII.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table IX shall be applied separately to each assembly in the sample.

4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subject to the provisions of MIL-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table VIII and only the test of item 307 in table IX shall be applied.

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4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table VII and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging,packing and container marking shall be in accordance with the provisions of packaging data sheet P11732360.

4.6 Methods of inspection.

4.6.1 Test equipment. Unless the assembly is procured apart from the next higher assembly, the test equipment in table X shall be used where specified to perform required tests.

4.6.1.1 AC voltage measurement. Unless otherwise specified, all ac voltage measurements shall be the in-phase component of the fundamental where MR1 (item 3.1 of table 1) is the phase reference.

TABLE X. Test equipment.

Item	Part Number	Nomenclature
1	11750202	Automatic Card and Unit, Test Set
2	11750422	Adapter, Computer Board - A5
3	11821457	Fixture Shock, Vertical and Horizontal Plane
4	11821458	Shock Adapter, Plate Vib Horz-Vert Plane/Shock-Z
5	11821500	Vibration/Shock Test Fixture - Computer Unit, FAT
6	11821520	Automatic Test Assembly Central Processor Unit
7	11821854	Test Fixture Auto Card, Test Set, Computer, FAT

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4.6.2 Performance tests.

4.6.2.1 R_0(AC) output. Use items 1, 2 and 6 of table X or suitable equipment to verify the requirements of 3.3.1.

4.6.2.2 R_0 $\overline{P}$ output. Use items 1, 2 and 6 of table X or suitable equipment to verify the requirements of 3.3.2.

4.6.2.3 R_0 $\overline{TA}$ output. Use items 1, 2 and 6 of table X or suitable equipment to verify the requirements of 3.3.3.

4.6.2.4 R_0 $\overline{WEAR}$ output. Use items 1, 2 and 6 of table X or suitable equipment to verify the requirements of 3.3.4.

4.6.2.5 Epsilon output. Use items 1, 2 and 6 of table X or suitable equipment to verify the requirements of 3.3.5.

4.6.2.6 Eta output. Use items 1, 2 and 6 of table X or suitable equipment to verify the requirements of 3.3.6.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 3, 4 and 5 of table X or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 $\pm$ 4 g with a time duration of 18 $\pm$ 3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table VII.

4.6.3.1.2 High intensity. Use items 3, 4 and 5 of table X or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 $\pm$ 10.0 g with a time duration of 1.5 $\pm$ 0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table VII.

4.6.3.2 Vibration. Vibration A as specified in 4.6.3.2.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.3.2.2 shall be conducted only for quality conformance inspection.

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4.6.3.2.1 Vibration A. Use items 4 and 5 of table X or mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2 procedure VIII, except that the test level shall be the vibration curve shown in figure 2 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table VII.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Use item 7 of table X or suitable equipment and subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operatng temperature shall be 140°F. The performance tests of table VII shall be applied.

4.6.3.4 Low temperature. Use item 7 of table X or suitable equipment and subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table VII shall be applied.

4.6.3.5 Temperature cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 3. At the conclusion of this test, subject the assembly to the performance tests of table VII.

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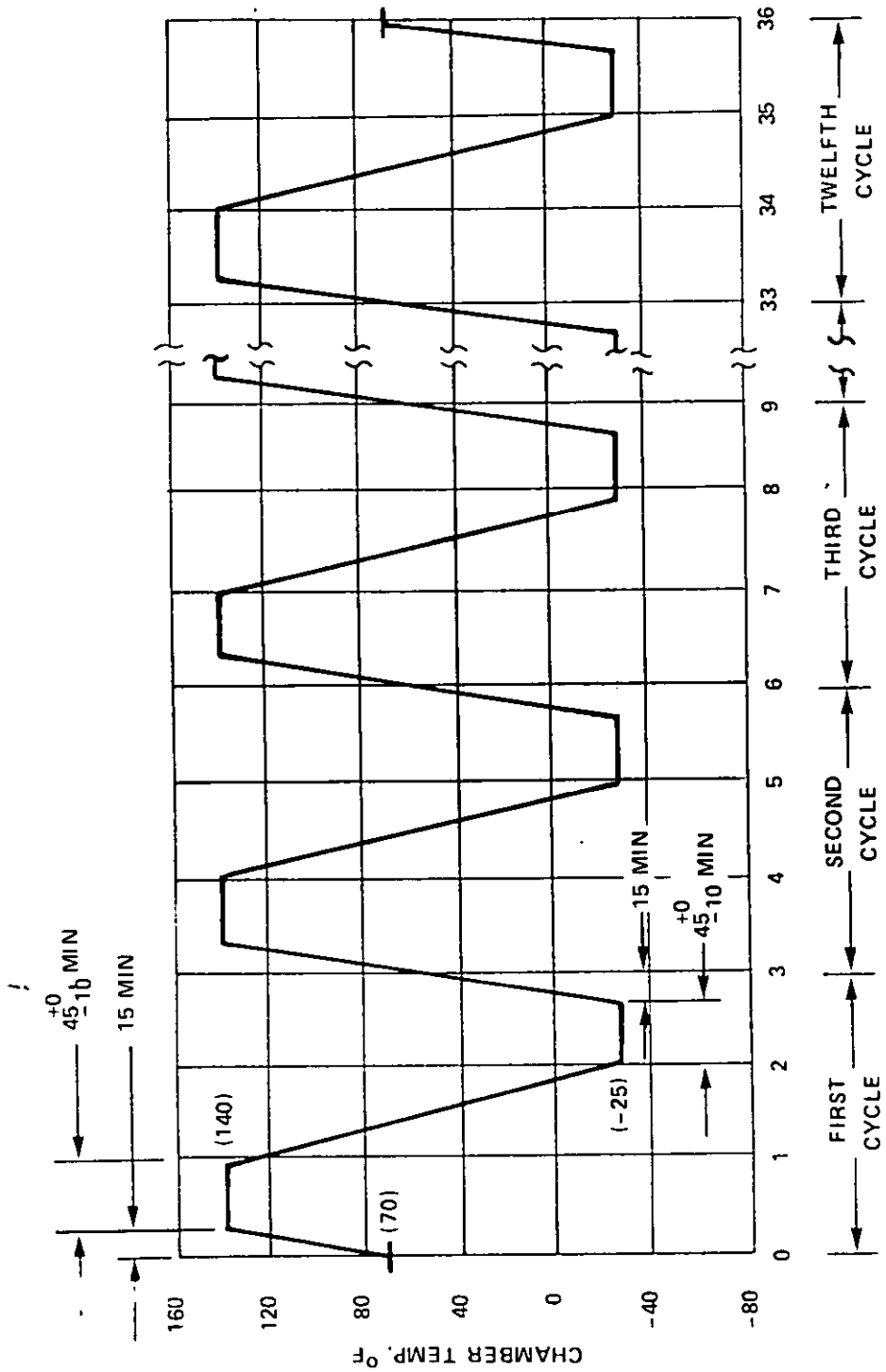


FIGURE 3. Temperature cycling.

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5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11732360. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The Superelevation and Drift circuit card assembly is used in the M21 Ballistic Computer Unit.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Special tests and examination as required, the type of inspection required, and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-C-48472B was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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